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Project Title:

Design and Validation of a Modular Shock Tube System for Transient Pressure Testing

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Introduction and Background

Design Problem, Motivation, and Need

In modern day military combat environments, it is commonplace for soldiers to encounter blasts and shocks from an array of devices such as grenades, IEDs, or other explosives. Even when a soldier is not directly affected by fragments from the explosive they are subjected to blast overpressure which can cause damage to equipment or contribute to injuries. To improve the protective systems for soldiers it is necessary to test and evaluate materials under controlled and repeatable conditions.

This is currently a problem that the U.S Army (DIU) is attempting to solve. They have recognized that field testing can be expensive, complex, and unrepeatable due to the unpredictability of explosives in the field. DIU has tasked the team with designing a modular shock tube to test a multitude of different materials in a controlled and repeatable environment. Currently DIU has a shock tube, but it has several issues that make it difficult to use and apply to their current applications. The shock tube is 30+ years old, it lacks modularity for their use, lacks sensors for material testing, and it's cumbersome and time consuming to use. As a result, DIU has a need for a newly developed shock tube that addresses these issues and their own requirements.

Intended use, users, and Operating Environment

The product's intended use is to expose new materials or protective design concepts to a controlled shockwave and to measure their durability and protective capabilities quantitatively and qualitatively. The primary users of the shock tube are going to be the army engineers, scientists, and technicians responsible for the design and creation of these new technologies. Analysts will be secondary users to interpret sensor data and compare different materials and designs. The shock tubes operating environment is going to be their facility in Alabama used for testing. This system is going to have its own area to be mounted down, also with its own access to pressurized gases.

Desired Functions, Features, and Interfaces

This project is purely mechanical and does not require software interfaces to be designed or built. At the start of this project the team was presented with 6 main requirements from DIU, however, as the project continued the requirements were refined into the following:

- The shock tube must **generate repeatable pressure shockwaves** within a range of pressures varying from **5-2000psi**
- The system must **fit within a 12'x8'** footprint

- **Cycle time <20min**
- The system must **abide** by **ASME** and **pressure vessel** standards

Because there is a heavy emphasis on modularity within the project some special features may include: configurable specimen attachments, configurable pressure differences, and standardized sensor locations. The shock tube will contain multiple interfaces for users: where they will change the pressure for each shock, where they will replace or exchange diaphragms, and where they will access the testing region to exchange materials. The shock tube will also have to interface with pressure regulators, valves, air compressors, sensors, materials, pressure gauges, and the general environment.

Value Statement

The proposed shock tube enables faster and more repeatable material testing. For stakeholders the new system provides reduced uncertainty in results which improves confidence in comparisons of materials. The system's modularity allows for faster cycle times and a wider range of testing. Ultimately the largest value of the shock tube is developing these materials to improve soldier safety in the battlefield, having the potential to **save lives**.

Additionally, this project is meant as a design and research project, where there will not be a prototype mainly due to safety and time considerations. **The intended purpose** of this project and report is to create a starting place for DIU engineers to learn about shock tubes, what details should be thought about throughout the design process, and potential challenges that they will run into.

Technical Issues, Challenges, Opportunities and Standards

There are a few challenges that the team has encountered so far in the design process. The first is reaching the 2000psi threshold; there are a lot of ways to reach this kind of pressure, however, some are very expensive and others decrease the safety for the users. Another challenge is designing a tube that will safely contain the gas at such great pressures while also allowing for fast cycle times and replacement of parts such as the diaphragm. These added features increase efficiency in testing but also make it a challenge to design for the safety required. The team thinks that the cycle time will also be a great challenge to solve because if a diaphragm is used the tube will need to be cleaned after each test. Depending on the material it can burst into many small fragments that can alter the wall effects of the tube leading to invalid and unpredictable results from testing. This cleaning will require the removal of those fragments, and if that section of the tube is not easily accessible then it will prove a great challenge to meet the 20min cycle time.

From a safety standpoint these high-pressure systems are mostly governed by pressure vessels and piping standards, while normal ASME standards still apply at various places. Our project will require a documented design basis that identifies the applicable codes, and defines design requirements, inspection, fabrication, and maintenance. This system will be treated as a high energy pressure system and will be designed with code informed practices.

Performance Metrics and Proof of Concept

To evaluate success of the system, this project will focus on measurable performance metrics tied to shockwave generation and system use. Key metrics include:

- Peak Pressure
- Pressure rise time
- Estimated Cycle time
- Size constraints
- System integrity and lifespan
- And reflection wave absorption percentage

Proof of concept will be established with:

- Design requirement documentation: translating client needs into engineering requirements and verifying that each requirement is addressed in design
- Safety and Compliance plan: documentation of hazard controls, documentation of stress and safety factor calculations and models for tube sections and all interfaces, through FEA
- System configuration: CAD based models of entire design concept including interfaces, sensor ports, driver, driven and testing sections, component interfacing and assembly details

Document Structure

The remainder of this report first explores existing shock tube solutions and relevant prior designs to establish baseline approaches and to identify challenges for the team's ideation. Building on that foundation, this report summarizes applicable codes and standards and evaluates their implications for a system operating at the project's target conditions. It then captures the client's needs and translates them into engineering requirements to guide design conception and ensure alignment with project objectives.

Next, the report presents market and stakeholder research to contextualize the potential impact and usefulness of the proposed shock tube system. It documents the team's design ideation and progress from project inception to now, as well as providing relevant engineering analyses for proof of concept. Lastly this report summarizes prototyping requirements and societal and environmental concerns for this project before addressing future steps and upcoming deliverables.

Existing Products, Prior Art and Applicable Patents

Overview of Current Products and Procedures

Shock tubes are most widely used for research applications to generate repeatable shockwaves for studying blast loading on structures, sensors, or materials samples. In practice testing procedures usually define a pressure time environment (peak pressure, duration of test, and the impulse), they specify hardware needed for measuring and evaluate repeatability of trials. In research facilities and/or labs a general procedure for using a shock tube involves setting a target pressure, selecting a testing material, and choosing measurement sensors [8].

Existing Systems in Use

Due to the nature of our project, that being in the field of research, there isn't much competition, however there are a lot of varied designs out there. All shock tubes work generally the same way. As shown in Figure 1, they have a driver section that builds pressure, a membrane to separate sections and hold pressure, a driven section that is held at atmospheric pressure, and a testing region. Where these designs vary come in the smaller sections once they are broken down.

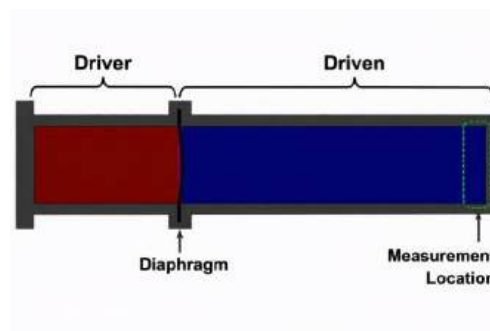


Figure 1 Standard Shock Tube Diagram

How the underlying Technology Works

A shock tube works by generating a pressure wave that propagates through a tube towards a testing region. This wave is generated by a pressure differential between the driver and driven section, then this pressure is rapidly released. The way this pressure is released can happen in multiple ways; the most common way is by diaphragm rupture. Once the pressure is released the high-pressure travels very quickly towards the low-pressure region.

The shock then propagates down the tube where it is measured in the testing section by a multitude of different sensors needed for material tests. Some parameters that are closely controlled in shock tubes are: the pressure in the driver section, the gas that is chosen for the tube, the geometry of the tube, the pressure release mechanism, and any venting that is done after testing [8].

There are multiple methods for initiating the pressure release in a shock tube; however, several commonly considered approaches introduce tradeoffs that can reduce system performance or increase design complexity. For example, valve release mechanisms may open too slowly or provide insufficient flow area to generate the desired shock strength, while piston concepts often require a tightly controlled interlocking and sealing mechanism that significantly increases mechanical complexity. The most used release mechanism is a diaphragm. The diaphragm acts as a sacrificial barrier that needs to be replaced after each use as it is ruptured in the event of testing. Diaphragm selection can be quite a challenge as they are directly impacted by driver pressure ranges. The diaphragm also has a strong effect on bursting pressure, rupture repeatability, and shock strength [9].

Prior Art and Designs

Conical Shock Tubes

Conical Shock tubes like in Figure 2 are often used to emulate spherical like explosion characteristics more effectively than constant area geometries like shown in Figure 1. As depicted in research study [10] it was determined that a cone angle of 38° creating a resulting blast wave that was near spherical. The authors cite that this wave geometry is due to the consistent interaction with the conical tube shape. This interaction has the capability to produce pulsations, in the rarefaction phase.

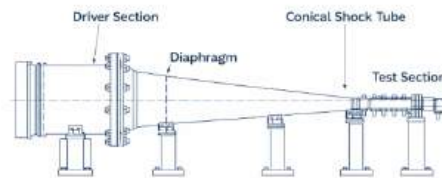


Figure 2 Conical Shock Tube Diagram

Convergent Shock Tubes

Convergent shock tubes as depicted in Figure 3 are very similar to conical ones but there is one key difference. The convergent tubes use a series of concave, convex, and oblique sections to shrink the tube diameter and create a perfectly planar shockwave that propagates smoothly to the testing material. This planar behavior allows for better controllability of the shockwave, and it also increases the power of the wave itself. Convergent tubes aren't as widely available industrially and produce a complex manufacturing process to achieve desired results [11].

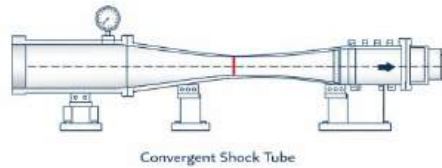


Figure 3 Convergent Shock Tube Diagram

Pneumatic Shock Tubes

Pneumatic shock tubes generate shock waves through compressing a gas and using a fast release mechanism. Basically, a gas builds up and is rapidly released to fire a piston that forces a lot of gas through the shock tube and that's the shockwave. These systems are widely used in labs as they are hyper modular. It's very easy to change the input pressure and they provide the most consistent shockwave out of most solutions. These tubes come with a huge drawback being that they are only useful for low pressures below 100psi [12].

Codes and Standards

In the case of a shock tube that holds and operates on pressures well above 15 psi, 500-2000 psi, it is classified as a pressure vessel. There are multiple existing codes and standards regarding the design, material choice, operation, and maintenance of pressure vessels from different established groups.

ASME Boiler and Pressure Vessel Code (BPVC), Section VIII holds the standards established for pressure vessel design. Division 1 of Section VIII focuses on the general design of pressure vessels. This establishes a general process for the development of a pressure vessel to ensure that it can hold the designate pressure of up to 2000 psi. Division 2 of section VIII goes into more detail on the analysis required for higher stress operations, most appropriate for the shock tube application. It provides the design requirements appropriate for high stress operating pressure vessels while also going into in-depth details on the associated analysis.

To determine the appropriate material that goes into the build of the shock tube, ASTM A240 was referenced. A240 provides a list of materials that were not only most used in pressure vessels but also rated to hold the desired pressure value. From the list of materials provided, AISI 304 was selected for its strength to withstand the holding pressure, corrosion resistance to the different applied gases, high strength-to-weight ratio, and excellent heat resistance to resist an instantaneous temperature rise caused by high pressure.

Operation code and standards for high pressure vessels can be found in OSHA 29 CFR. Subpart H describes codes associated with hazardous material. Due to the hazards that highly compressed gases, including air, pose, it is necessary to be familiar with this subpart during shock tube operation to understand the procedure required for handling the gases. Subpart M details the code associated with the use of compressed air and compressed air equipment. Subpart O describes machine safety, specifically relating to machinery and machine guarding.

For inspection, maintenance, and in-service evaluation, API provides codes and standards regarding the procedure required for high pressure vessels. Section 510 and 572 both go into detail on pressure vessel maintenance that need to be referenced to not only extend the life of the shock tube but also ensure failure due to fatigue is not reached. 510 formally describes Maintenance Inspection, Rating, Repair, and Alteration pertaining to pressure vessels. While 572 is labeled Inspection of Pressure Vessels.

Customer Requirements and Engineering Design Specifications

Customer Requirements

Through communication within the team and with the project sponsor, a list of client requirements were established then organized into various engineering requirements. The most critical requirements stem from generating a shockwave at a range of different pressures while following safety specifications put in place by multiple institutions. Some other requirements feature size constraints, fast cycle times, reflection wave reduction, and industrially available parts. The house of quality, Figure 29, quantitatively confirms the most important specifications being pressure capabilities and safety within the system.

Functions and Performance Metrics

A function tree, Figure 30, was created to understand all the subfunctions of the shock tube. At a high level the system needs to store pressure, release that pressure, deliver that wave to the testing section, observe the effects and test, and return the system to a safer

state through venting. These high-level functions come with multiple sub functions that provide design opportunities.

Because the project is being developed for material testing under blast conditions the performance metrics focus on system capabilities. These metrics include: a pressure range of 5-2000psi, cycle time less than 20 minutes, target vent time less than 10 seconds, and a pressure set time of less than 30 seconds. These metrics describe the system as a whole and how it functions through multiple tests. Some other metrics include a sound target less than 85db as that's where last damage can take place in ears, a window size of 2" radius, a 50% reduction in reflected shockwave strength. Some of these metrics will be refined as the design process progresses and more information and modeling is done.

However, since the inception of this project the sponsor and the team have cut some of these metrics from the project as they are either unimportant, achieved without design effort, or too time consuming for the time allotted. For example, the shock tube will not have a viewing window because, as noted by the sponsor, it's not entirely relevant as the test pieces can be extracted after testing. The tube also inherently reduces the reflected shockwave by more than 50% without any design efforts, it was also noted in a sponsor meeting that the team shouldn't worry about this requirement. The team also dropped the noise target as there is not a reasonable way to test this without a prototype, prior research did not uncover any details pertaining to the sound level of a shock tube.

Constraints

For this system there are defined geometric constraints that the system has to fit within a 12'x8' area. In addition, the design must be compatible with DIU's lab infrastructure. There are also basic manufacturing constraints that also apply, the team has set a metric that 50% of the components used in the system must be commercially or industrially available. This constraint will help simplify the process and reduce the need for models and calculation as these products should already abide by engineering standards. The last constraint is all the design and engineering standards outlined in the Codes and Standards section of this report.

Market Research and Potential Impact

Market research for this project focused on understanding the operational need, health impact, and validation gaps related to blast overpressure exposure in military environments. Information was gathered from Department of Defense policy documents, U.S. Army publications, and peer-reviewed research on blast injury mechanisms, sensor accuracy, shock tube testing, and mitigation effectiveness. These sources reflect the

perspectives of key stakeholders, including military leadership, engineers, medical researchers, and end users responsible for Soldier safety and readiness.

Recent DoD guidance establishes an interim blast overpressure safety threshold of approximately 4 psi, making blast exposure management a current operational requirement rather than a purely academic concern. Army publications further emphasize that injury risk increases as overpressure levels rise, reinforcing the need for reliable methods to evaluate and control exposure during training and operations. Health-focused studies show that repeated blast exposure is associated with neurological and cognitive effects even in the absence of visible injury, directly affecting long-term Soldier health and force readiness if not properly addressed.

At the same time, prior studies evaluating body-worn blast sensors demonstrate that measured pressures can vary significantly based on sensor orientation, placement, and signal processing. This variability highlights the limitations of relying solely on field data and underscores the need for a controlled, repeatable test platform to validate sensor performance. The literature also shows that shock tubes are an accepted and scalable method for generating blast-like pressure transients, making them well suited for calibration, validation, and comparative testing of sensors and mitigation concepts.

Research on blast mitigation further indicates that materials and interface geometry can meaningfully alter transmitted and reflected pressures, supporting the feasibility of studying pressure-reduction strategies in a laboratory setting. These findings directly influenced the design emphasis on tunability, repeatability, and the inclusion of a dedicated reflection test section with a target of significantly reducing reflected pressure relative to a rigid surface.

The primary impact of this project is expected in terms of time, cost, and development efficiency for the sponsor. A modular shock tube system with a cycle time under 20 minutes could support approximately 20–25 tests per day, compared to only a few tests per day using larger-scale or field-based methods. This increased throughput can save dozens of engineering labor hours per week by reducing setup time, minimizing ambiguous results, and limiting the need for repeated testing. By improving sensor validation and enabling systematic evaluation of mitigation concepts, the system also supports earlier identification of design issues, reducing downstream redesign costs and contributing to long-term reductions in health and readiness burdens associated with blast exposure.

Design Concept Ideation

For the ideation process, function trees/ decomposition, train of thought ideation, customer journey maps, and a few other tools were used to delve into how the user is going

to experience the product, how the product will function, and how the product will be made.

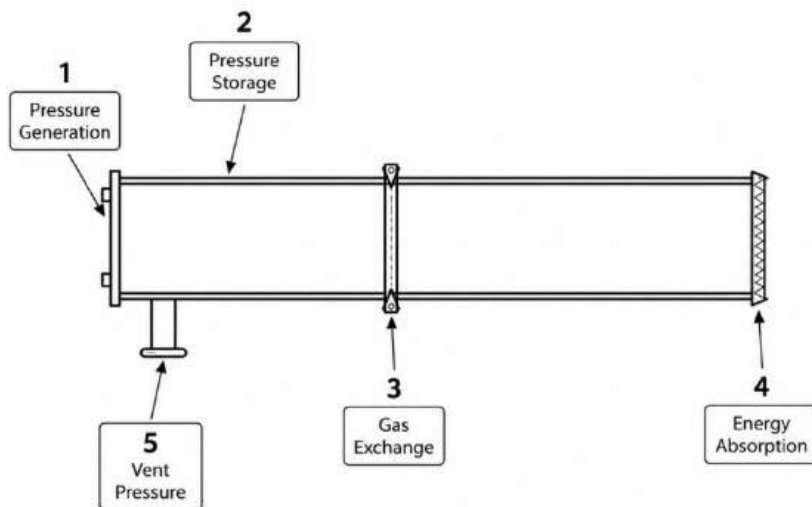


Figure 4 Sketch of basic shock tube decomposed into sub-functions

First, a basic depiction of a full shock tube as seen in the above Figure 4 was created. Then ideas were generated for each of the function sections. For this part of the ideation, it was important to broaden the scope to come up with ideas that are out of the box and that are sometimes implausible so that more creative solutions could be created. Titling each page was a specific subfunction needed for the generation of shock, below it anything we could think of that might do that function was written down. The pages were used as a starting point for research.

After narrowing the scope, three different preliminary concepts were created which could be used to generate the required shock waves for testing the sensors and materials. The first is based on a traditional shock tube and is simple to manufacture, simple to do analysis on, and doesn't require many specialized parts (Figure 5). The biggest issue with this concept is that it requires the generation and storage of extremely high pressures which pose high risk to those in the immediate vicinity.

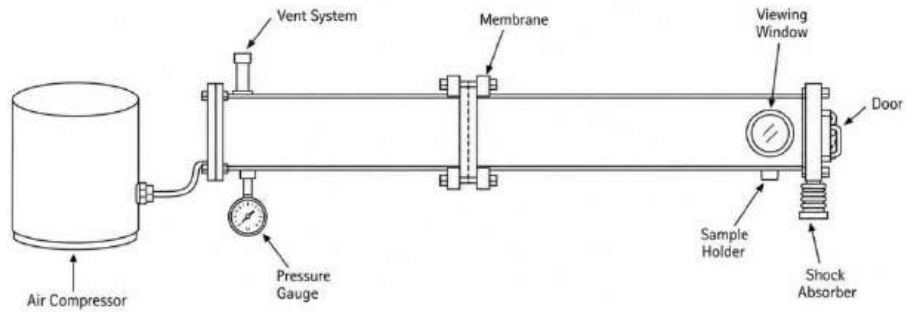


Figure 5 Concept 1 preliminary design sketch using basic shock tube design

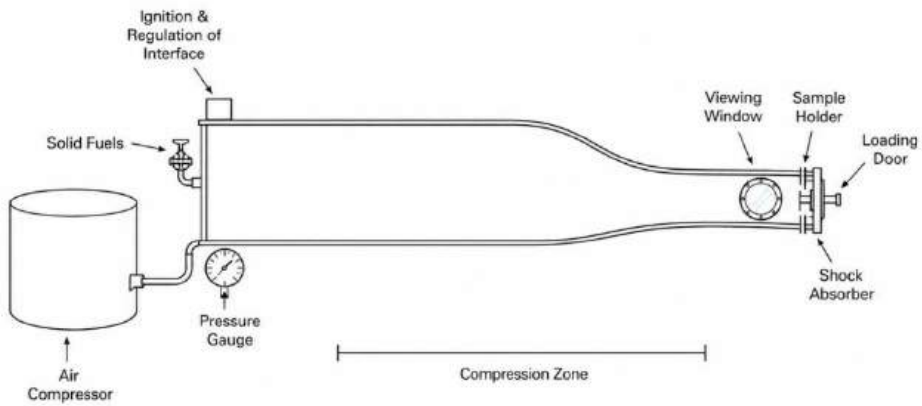


Figure 6 Concept 2 preliminary design sketch using solid fuel and compression zone

The next concept as seen in Figure 6 uses solid fuel to generate the shockwave that propagates down the tube and impinges on the test specimen. One of the possible issues with this is that solid fuels burn in different ways each time and that variance in burn path and time can change the shock front. Secondly, when the fuel burns it releases fumes into the tube that can change the way the shockwave moves down the tube and can create turbulence in the flow that is otherwise not present in other solutions. Lastly, the burning of a fuel solid or otherwise in the tube will generate heat which changes the temperature and reduces the similarity to free field blasts. This method of generating a shockwave is good due to the high pressures achievable and the lack of need for a membrane due to the shockwave being generated like how it would in a free field scenario.

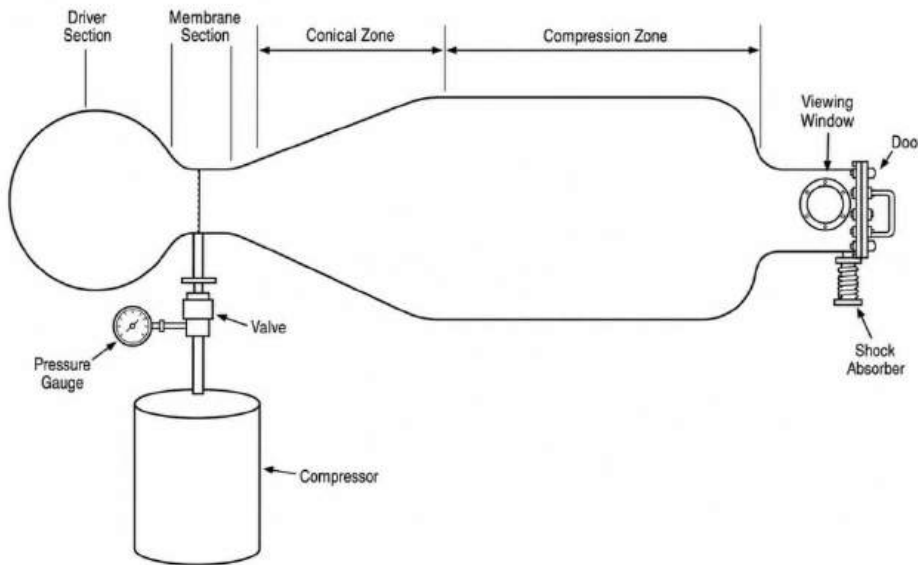


Figure 7 Concept 3 preliminary design sketch using compression zone, valve, and conical geometry

The last concept is to use a pneumatic system to release the shockwave into the tube. There are a couple big issues with this shock generation system (Figure 7). One is that achieving sub 100 microsecond release time for the volume requires highly specialized parts that are not readily available. The second issue is that finding a valve that operates at such high pressures is not accessible given the sub millisecond response requirement. For

the third concept, an alternative is creating a choke point allowing a smaller valve which can operate under the needed conditions. The second and third designs also use the geometry of the tube to their advantage by creating a tube with convex and then concave curves, the shockwave is reinforced and thus the pressure of the wave increases which means that the system doesn't require as high initial pressures as it would with a traditional shock tube. Another strength of the geometry shown in the second two concepts is that they both realign the shockwave so that it is more planar and thus there is a greater surface area that is useable for testing.

Concept Selection and Justification

Though there are drawbacks with each of these systems, we have chosen to move forward with concept one due to the large amount of literature about standard constant radius shock tubes, the reduced complexity of computational analysis, the increase in safety of this design. Concept 1 which consists of two tubes, one 3 ft long and one 8ft long connected by a collar which contains the membrane with end plates to cap off the design is much simpler computationally due to the constant diameter of the tubes. This reduction in computational complexity increases the ability to accurately calculate stresses in the system, fluid dynamics, and overall effectiveness. The complex geometries of concept 2 and 3 provided a few benefits, though these didn't outweigh confidence and the cost reduction provided by concept one. Additionally, time constraints of the semester being only 4 months there is not enough time to fully consider the more complicated designs, even if they offer benefits with the requirements and desires of DIU. The design is safer due to the reduction in complex curves and angles which act as stress concentrators, additionally there are far more regulations and guidelines that can be followed to design a safe machine that will provide reliable results.

The team's **recommendation** is that concept 1 is the easiest, most documented and researched method for designing and building a shock tube. However, alternate geometry tubes should not be discarded immediately depending on desired project length.

Shock Tube Component Design

Main Tube Design

The main tube design consists of two different sections, the driving section and the driven section. The exploded assembly is given in Figure 8.

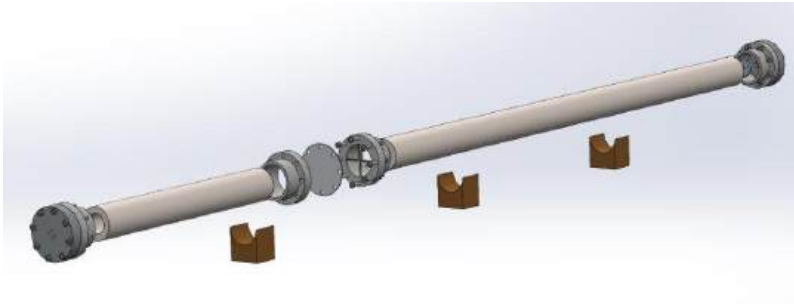


Figure 8 Exploded Full Tube Assembly

There are a couple important design considerations when developing the dimensions and material choices for the tube. Each component will be given its own section in which the mathematics and design equations will be written out, and each step will be explained. The components in order of the flow of the gas within the system are; the intake system/ exhaust system, end wall, driving section, collar flange interface, membrane, driven section, testing material mounting plate and opposing end wall. The components of the intake and exhaust system are chosen based on the necessary flow characteristics with some thought put on their acoustic characteristics. The rest of the design maximizes the safety and efficient shock wave generation. The fully assembled design will provide repeatable, precise, and accurate shockwaves for material and sensor testing.

Diaphragm Design

The diaphragm assembly is positioned at the interface between the driver and driven sections, clamped between the two bolted flange collars. Its function is to maintain a pressure seal between the high-pressure driver gas and the atmospheric-pressure driven section until the target burst pressure is reached.

The diaphragm itself is a steel plate. Steel was selected for its predictable and consistent rupture behavior, low cost, and the ease with which thickness can be varied to tune the burst pressure. By changing diaphragm thickness, the system can be configured to operate across the full 5 to 2000 psi pressure range without any modification to the tube assembly itself.

On the driven side of the diaphragm, a cross-blade knife is mounted facing the diaphragm surface. The knife consists of two perpendicular blades forming a cross pattern across the inner bore diameter. As driver pressure is increased, the diaphragm deflects forward under the pressure differential. When the pressure reaches the burst threshold, the diaphragm contacts the knife blades and ruptures along the four quadrants defined by the cross

pattern. This produces a clean, symmetric four-petal rupture rather than a random tear, which is critical for two reasons. First, the petals fold back cleanly against the tube wall without obstructing the flow path or generating debris that could interfere with shock wave propagation or test instrumentation. Second, the symmetric rupture produces a planar, well-defined shock wave front rather than an asymmetric pressure pulse, which is essential for repeatable and consistent test results across shots [19].

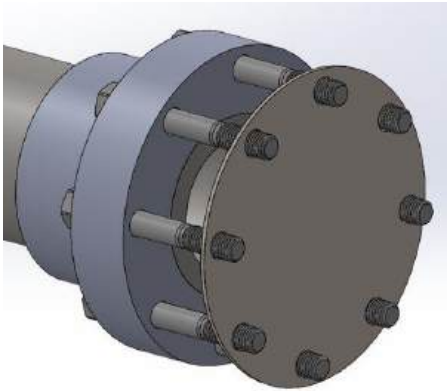


Figure 9 Diaphragm Design

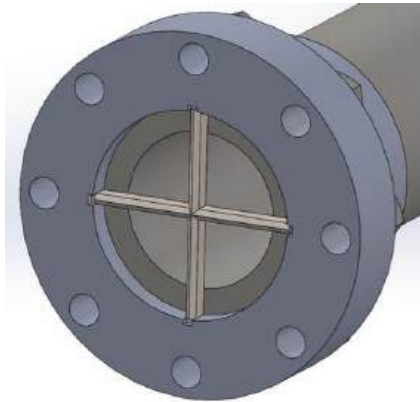


Figure 10 Diaphragm Blade Design



Figure 11 Diaphragm Blade Design Inspiration

Driver and Driven Section Coupler Design

The driver and driven sections are coupled at the diaphragm interface using a dual bolted flange collar assembly. The flange collars are welded to the ends of each tube section and bolted together to form a pressure-tight joint. The coupler serves two functions: it clamps

the aluminum diaphragm between the two mating faces, and it transfers the axial pressure load from the driver section across the joint to the driven section.

The flange collar geometry was designed around the tube outer diameter of 5 inches with a bolt circle diameter (PCD) of 177.8 mm (7 inches). Eight bolt holes are equally spaced around the bolt circle. The flange face provides a seating surface for the diaphragm plate, and the bolted connection applies the clamping force necessary to maintain the pressure seal at the interface [20].

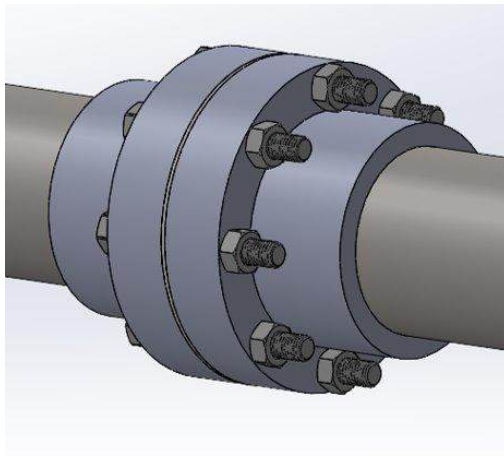


Figure 12 Collar Design Between Driver and Driven Section

Air Intake and Exhaust System Design

The air intake sub-system is a serial multistage pipeline that connects a gas reservoir to the driver section of the shock tube. Figure 13 showcases all components that go into pressurizing the gas, containing the high-pressured gas, and the depositing process into the driver section.



Figure 13 Layout of the shock tube intake system that deals with driver section pressurization.

The sub-system consists of a tank that houses the testing gas at atmospheric pressure to be used in the driver section for the shockwave test. There is an isolation valve used to shut off gas flow through the rest of the system meant for maintenance, operation, and emergency purposes. The Gas Booster is used to pressurize the testing gas from the reservoir to desired operation pressures. Check valves are present to regulate the direction of flow, preventing high pressure gas from flowing backwards into low pressure areas during operation. Though most gas boosters have internal check valves to prevent the backflow of high-pressured gases, an additional safety precaution is included to prevent damage to the gas booster and safety reasons. Pressurized gas will be stored in a 3-Cylinder Cascade subsystem at different pressures, high to low, for constant high flow during filling. A gauge and pressure relief are included to ensure that the high-pressure cylinders are relieved of pressure when reaching a critical state. The high pressure regulator takes the erratic high-pressure gas from the cascading cylinder during filling and turns it into steady outlet pressure that is stable downstream and useable without major damage to equipment. The inclusion of a needle valve provides precise regulation of flow rates and high-pressure sealing that standard valves cannot achieve, enabling accurate throttling, metering, and shutdown during the filling process.

The gas booster recommended for this system is the **Haskel AG-50 Pneumatic Gas Booster** as shown in Figure 14. This booster can pressurize gas up to 39,000 psi, which far exceeds our pressure requirement in the 3-cylinder cascade. It is designed to handle N₂, He, CO₂, O₂, H₂, and Argon. Since it is an air-driven device, the risk of combustion is low even when these compounds are at high pressure. It also has fast cycling depending on the air supply.



Figure 14 Haskel AG-50 Pneumatic Gas Booster

The **Air Systems International 3-Cylinder Cascade Fill Assembly, 4500 PSI** as shown in Figure 15 is recommended. Unlike a single cylinder fill system, by having 3 cylinders at varying pressures in descending order, there will be a constant high flow during the filling. This sub-system option is also optimal for filling larger volumes at a faster filling rate.



Figure 15 Air Systems International 3-Cylinder Cascade Fill Assembly, 4500 PSI

The air exhaust sub-system is a parallel pipeline that connects the driver section to a pump and a waste line. The layout of the sub-system is shown in Figure 16. The isolation valve is present to control the access of the driver section to the exhaust subsystem

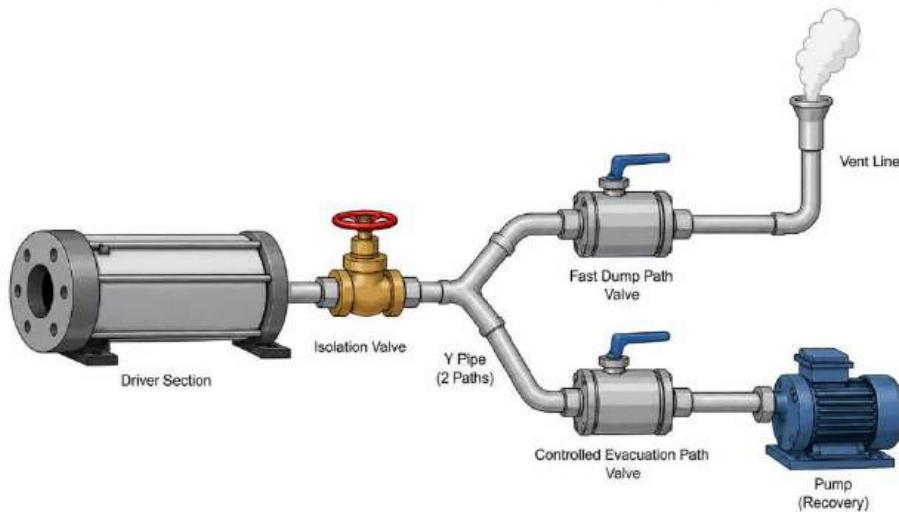


Figure 16 Layout of the shock tube exhaust system that handles test gas waste disposal and test environment baseline recovery

The isolation valve is present to control the access of the driver section to the exhaust subsystem. This device is mainly present to prevent an outflow of test gas during shock tube trials. Then follows a Y pipe that splits the pathing connection to the driver section into 2. Each path is blocked by a control valve that is not to be opened simultaneously. The fast dump path is present for initial fast exhaust of the test gas that was used during the trial. What follows the vent line depends on the method of disposal depending on what gas was used for testing, whether it be an ordinary vent for air or a container for more dangerous compounds. The other path to be used after the fast dump path is the controlled evacuation path. This connects a pump to the driver section for test environment baseline recovery. In the case there is a specific baseline pressure required for each test, this line is present to depressurize the driver section prior to gas intake. There is no recommendation for the pump in this sub-system as this can be any generic pump appropriate for intended operation.

Driven Section End Wall Design

The driven section end wall consists of a plate that is circular and has a specified thickness given later in the calculation section depending on the specified max pressure for the

system. The end wall has eight holes around its perimeter to fasten to the flanged collar of the driven section and then four holes to mount the testing mount plate Figure 17. Each of these stress concentrations are accounted for in the later calculations.

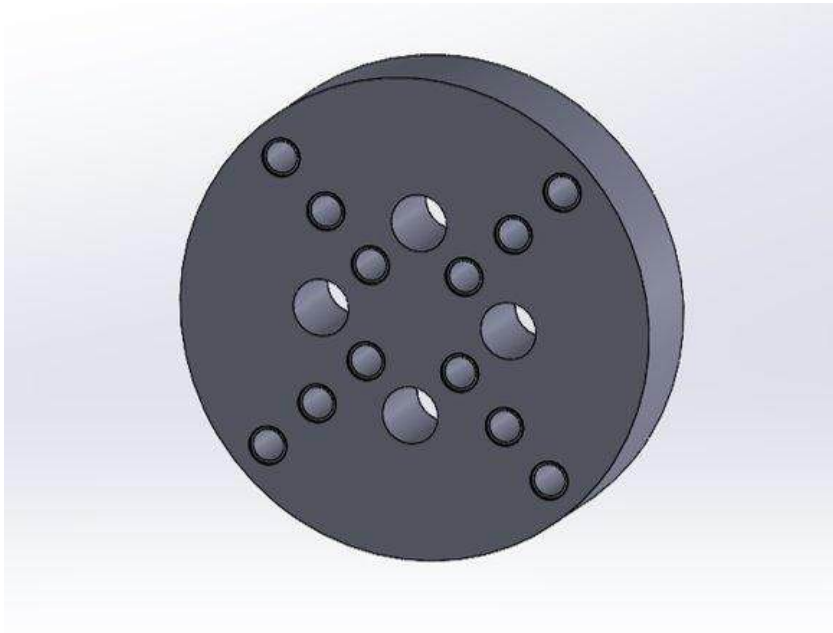


Figure 17 End Wall Mounting Plate CAD Model

Material Selection

Material selection for the shock tube assembly was guided by pressure rating, corrosion resistance, and structural compatibility between mating components.

AISI 304 stainless steel was selected for all pressure-bearing components: the driver section, driven section, flange collars, and end plate. AISI 304 provides a yield strength of 207 MPa (30,000 psi), good corrosion resistance, and is a widely used material for pressure vessel applications [21]. Using the same material across all pressure-bearing components eliminates galvanic corrosion at the flange interfaces and ensures uniform thermal expansion across the joint, preventing differential stress during pressurization cycles.

ASTM A36 structural steel was selected for the support brackets. The support structures carry no internal pressure load, they are loaded only in bearing and bending from the weight of the tube assembly. A36 provides a yield strength of 250 MPa, which is more than

sufficient for this application, and is significantly more cost-effective than stainless steel for non-pressure-bearing structural components [22].

Industrial Design

The main considerations for this design are in making sure that it is simple to use, quick, and safe. To design a product that is safe for human interaction indicators must be installed in the system so that the users know when they can interact with the product safely and when they should be weary or stay away from it. The product must also be biomechanically safe and efficient to interact with by ensuring that any lifting loads on the user are low and making sure that there is low cognitive drain from the system interface. When using the system as seen in the journey block diagram in Figure 12, in between tests you start by separating the driven section from the driving section, then you replace the membrane and re-attach the two halves then you set the pressure to begin the test. By sequentially ordering the tasks, the user is not overwhelmed, and they are able to make sure each task is completed thoroughly. After each task is completed, there is a visible marker such as the indicator on the bolts providing visual indications of correct preload or the light that is activated when the pressurization of the system begins. Another factor is in making the system quick to use. Elements such as the rolling track that the driven section is supported by and the easy controls to set the pressure provide this quick response. When rolling the driven section away from the driver section in between tests, the tracks have stops at the end so that the driven section which is approximately 200lbs is strongly constrained to where it is supposed to travel. Additionally, the components of the system that are sharp or that pose risk of pinching, are either within a housing or are marked for quick identification. These usability components of the design provide a system that is utilitarian and that is safe to use.

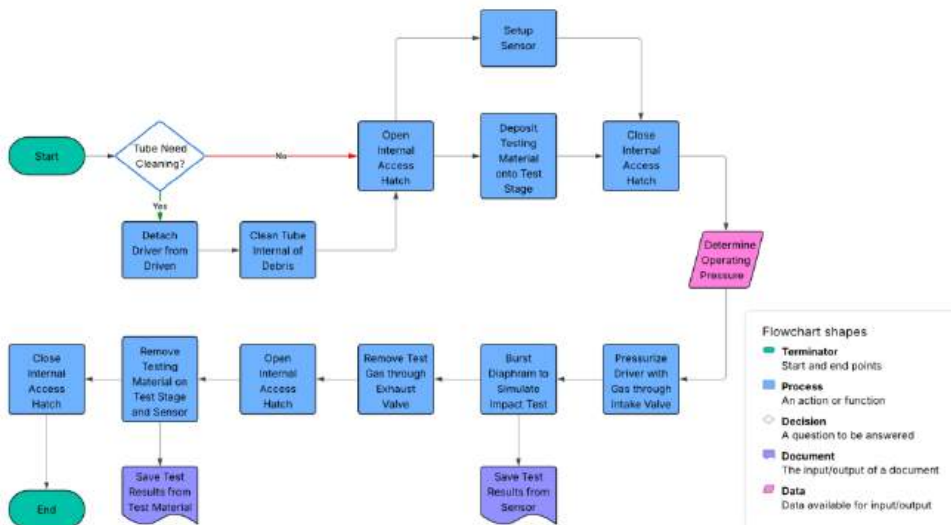


Figure 18 Block Diagram of Shock Tube System

Engineering Analyses and Experiments

Pressure Region Calculation

Pressure Regions in a Shock Tube

Shock tube flow is described using a numbering system for pressure regions that define the states of gases before and after diaphragm rupture. According to Figure 19 and Figure 20, there are 5 different regions within a shock tube. Before a test has started there is pressure in both the driver and the driven sections, and these are pressure regions 4 and 1 respectively. The driver region contains highly pressurized gas while the driven section contains gas usually at atmospheric pressure to create a high-pressure gradient during testing and increasing the shock front strength. Once the diaphragm is ruptured the stored pressure difference in the system drives an incident shock wave into the driven section creating expansion waves, an incident shock, and pressure regions 2 and 3. The expansion waves are created as the high pressure driver section decompresses into the driven section, these waves travel in the opposite direction than the shock front and can be seen in Figure 20. The incident shock is the shock front that will be used to test materials. Pressure region 2 is located right behind the incident shock, where pressure region 3 is located behind the contact surface. This contact surface is where the gas in the driver section and the gas in the driven section meet each other.

As the incident shock travels through the driven section it increases the pressure, density, and temperature of the driven gas from region 1 to region 2. When the incident shock hits the end wall of the tube it is reflected towards the driver section and creates region 5 at the testing location. This region has the second highest pressure within a shock test and is generally where testing materials are placed as this is where the loading will be the highest. All these pressure regions are vital for both understanding how to evaluate materials but for also understanding the loads on the system for tube design [23]

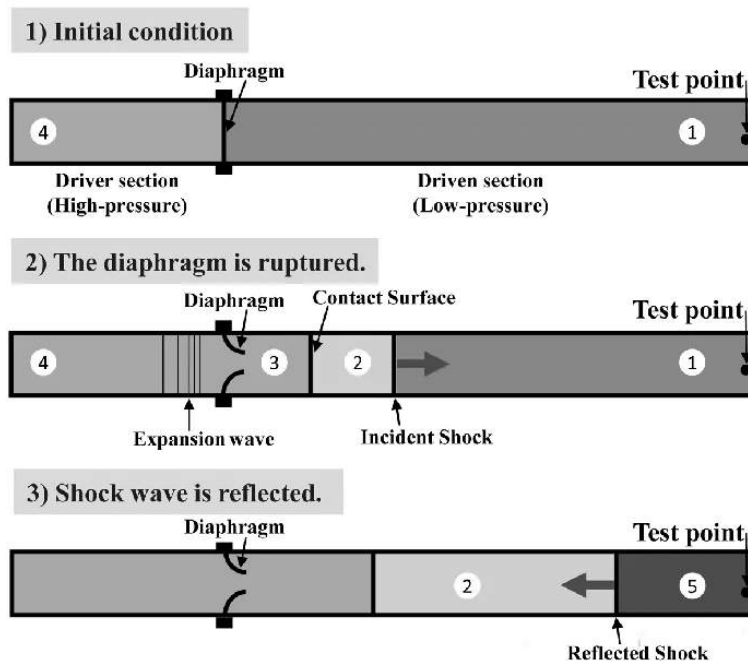


Figure 19 Shock Tube Pressure Regions

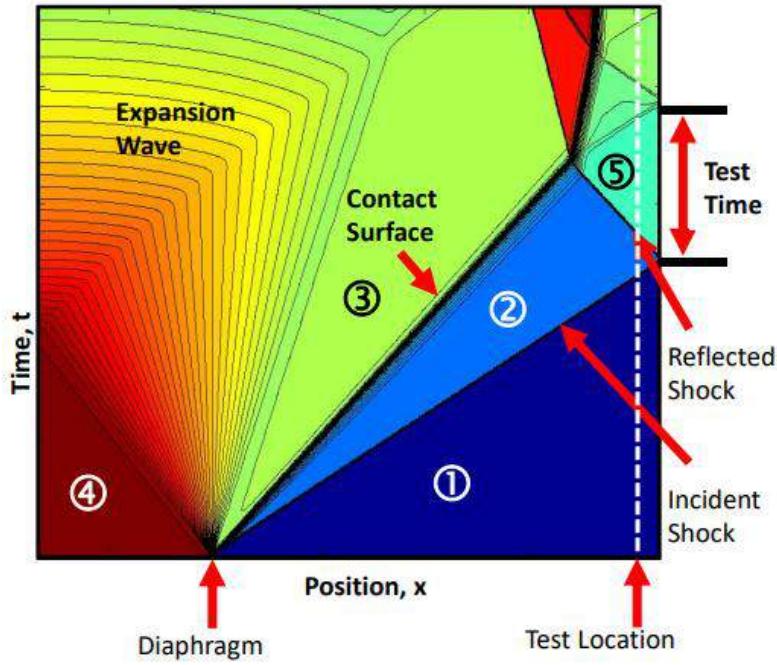


Figure 20 X-T diagram of shock wave test

Pressure Calculation Method

Pressure region calculations and gas combination graphs for this work were performed in MATLAB with an altered code provided by Dr. Mohammad Mohaghar from his thesis work with shock tube perturbation. Using his derived equations for calculating pressure regions the team was able to perform calculations for the pressure in different regions of the shock tube based on the gas combination used in the driver and driven sections, the initial system temperature, and the Mach number desired [24]. Results were compared with an existing shock tube region pressure calculator [25]. The following equations, 2 through 8, along with Table 1 provide a roadmap to the calculation method.

$$P_2 = P_1 \left[1 + \frac{2\gamma_1}{\gamma_1 + 1} (M_s^2 - 1) \right] \quad [1]$$

$$P_4 = P_2 \left[1 - \frac{(\gamma_4 - 1)(a_1/a_4)(P_2/P_1 - 1)}{\sqrt{2\gamma_1(2\gamma_1 + (\gamma_1 + 1)(P_2/P_1 - 1))}} \right]^{\frac{2\gamma_4}{\gamma_4 - 1}} \quad [2]$$

$$u_2 = \frac{2a_1}{\gamma_1 + 1} \left(\frac{M_s^2 - 1}{M_s} \right) \quad [3]$$

$$a_2 = \sqrt{\{\gamma_1 R_1 T_2\}} \quad [4]$$

$$M_2 = \frac{u_2}{a_2} \quad [5]$$

$$M_r = \frac{(\gamma_1 + 1)M_2 + \sqrt{(\gamma_1 + 1)^2 M_2^2 + 16}}{4} \quad [6]$$

$$P_5 = P_2 \left[1 + \frac{2\gamma_1}{\gamma_1 + 1} (M_r^2 - 1) \right] \quad [7]$$

Table 1 Definitions Variables used in Pressure Region Calculation Equations

Variable	Description
P₁	Initial driven pressure
P₂	Pressure behind the incident shock
P₄	Initial driver pressure
P₅	Pressure behind the reflected shock
Y₁	Specific heat ratio of the driven gas
Y₄	Specific heat ratio of the driver gas
a₁, a₄	Speeds of sound in the driven and driver gases
M_s	Incident shock Mach number
M_r	Reflected shock Mach number

The first step is to calculate the pressure in region 2, P₂, behind the incident shock using the initial driver pressure, P₁, the specific heat ratio of the driven gas and the incident shock Mach number. This equation determines how strongly the gas in the driven section is compressed during testing and is necessary to calculate the required pressure in the driver section, P₄. This pressure can be determined using the pressure ratio between P₂ and P₁, the speed of sound ratio of driven and driver gases, and their specific heat ratios. This equation is vital for the design process as this is how the team meets the client requirements and designs the tube for ASME safety standards.

Additionally, the shock pressure in the testing region after reflection, P5 can be found using P2 and the reflected shock Mach number. This pressure provides DIU with an understanding of what loads their materials will undergo during each test.

Gas Combination Screening

For shock tube testing there are many gases that are commonly used in the driver and driven sections. The gases and their combinations have various effects on the Mach number and pressures within the system. The gases that were analyzed for this system were air, argon, helium, CO₂, and nitrogen for both the driver and the driven sections. These gases were selected because of their wide range of molecular weights and abundance in shock tube testing. However, not all combinations of gases were considered practical for current designs. For example, helium as a driven gas consistently produced P4 values, pressures that the operators would need to fill the driver section up to, that were astronomically large, in the millions of psi, for Mach numbers as low as 1.6.

Interpretation of Gas Combination Heat Maps

To compare gas combinations, heat maps were created in MATLAB showing the required driver pressure, P4, for each driver and driven gas combination per Mach number of the system. The current design requirement for the system is 2000psi as a P4 pressure. For each heatmap below in Figure 21 through Figure 24, the Mach number listed in the title indicates the shock strength associated with that combination of gases. Figure 21 was included as a reference for what Mach number is required for 500psi for air and nitrogen configurations. The boxed gases are the driver and driven combinations that reach the required driver pressure for that Mach number.

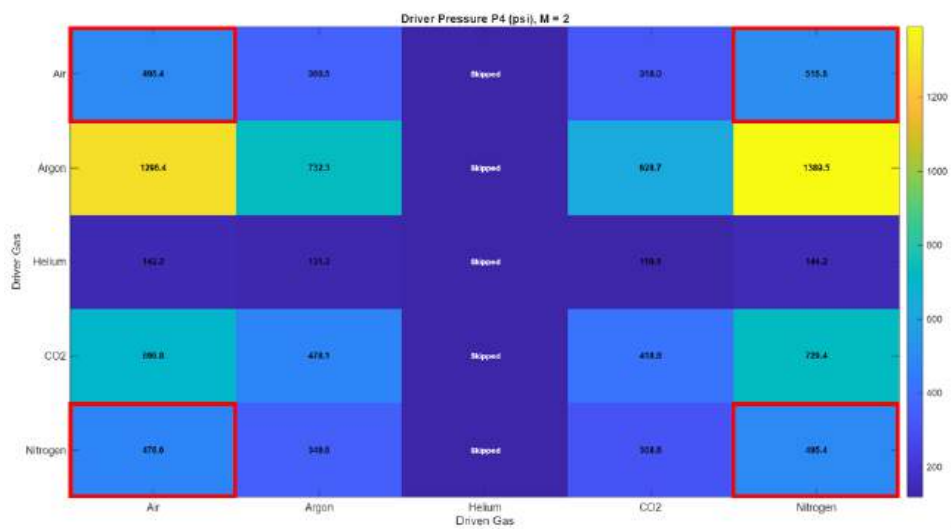


Figure 21 Driver Pressure Heatmap M = 2, Air and Air, Air and Nitrogen, Nitrogen and Nitrogen Combination

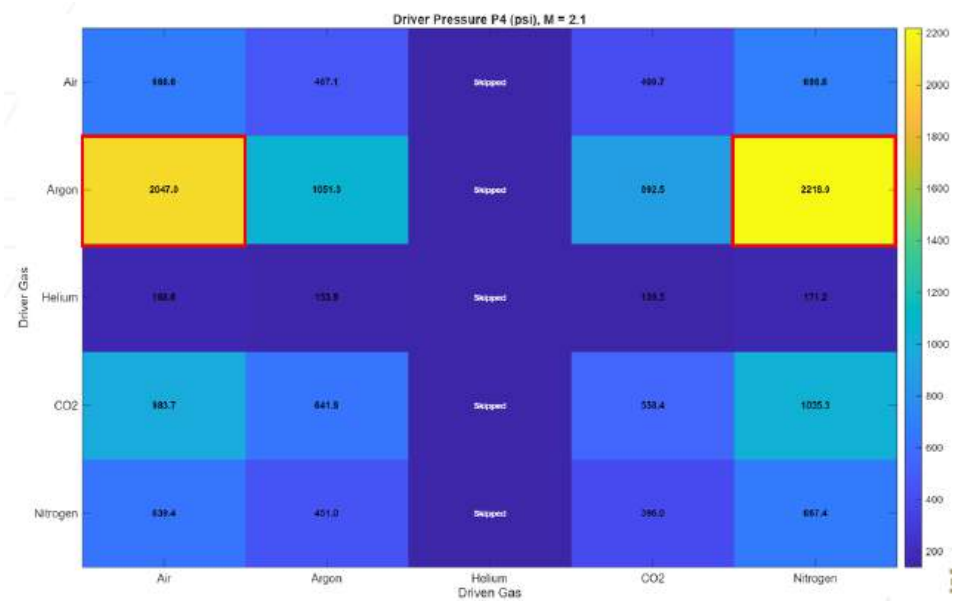


Figure 22 Driver Pressure Heatmap M = 2.1, Argon and Air, Argon and Nitrogen Combination

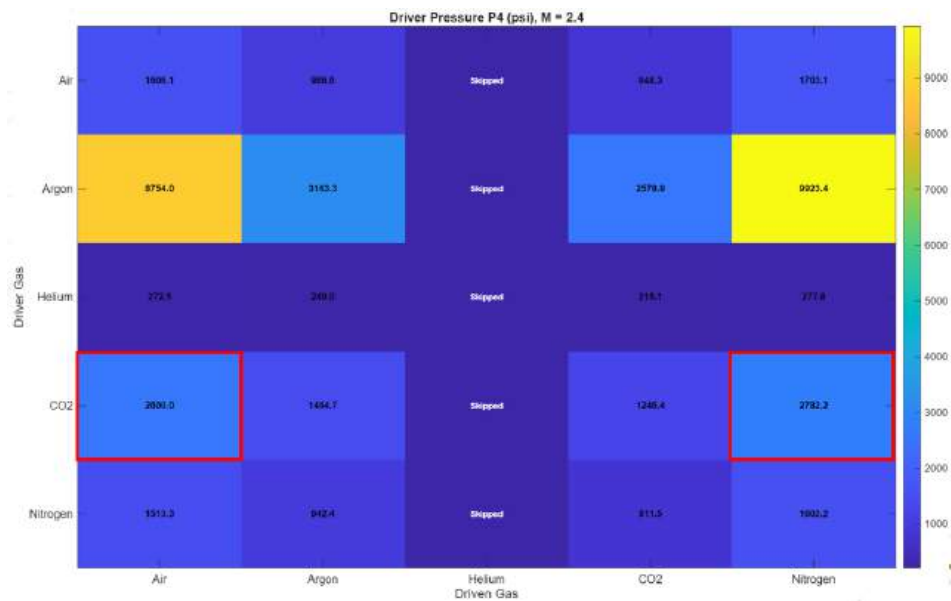


Figure 23 Driver Pressure Heatmap M = 2.4, CO2 and Air, CO2 and Nitrogen Combination

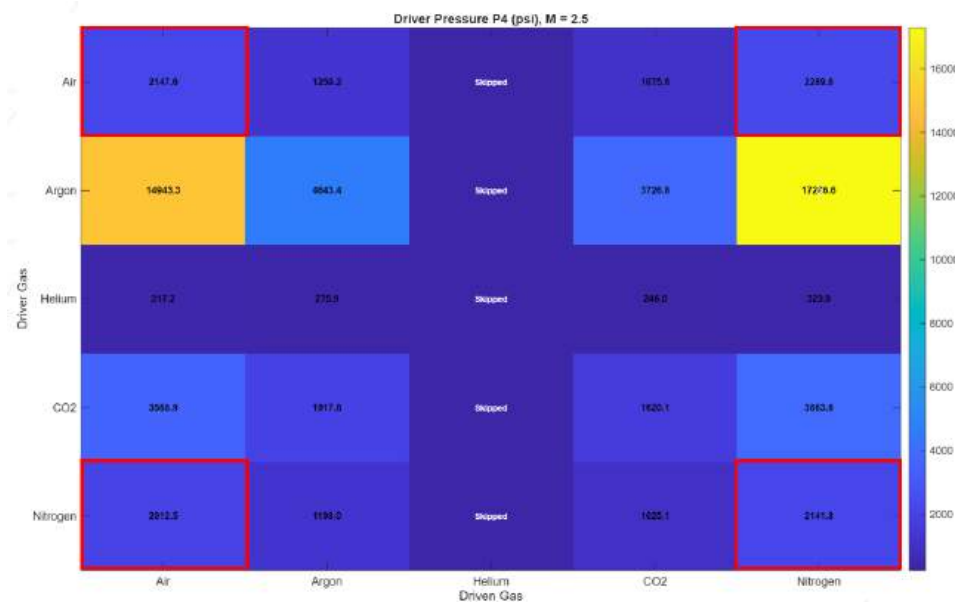


Figure 24 Driver Pressure Heatmap M = 2.5, Air and Air, Air and Nitrogen, Nitrogen and Nitrogen Combination

The figures above represent how many combinations and variables are included in shock tube testing. In other words, the plots show how different gas combinations can achieve the same structural design limit in the driver section while producing different incident shock Mach numbers.

The team has concluded that the **Mach number is the critical parameter** in shock tube testing because it governs the strength of the incident shockwave that the testing specimen experiences. Higher Mach numbers correspond to stronger shocks with more overpressure at the testing location. These results are shown in Figure 25. The gas screening above was a critical performance trade study because a gas combination may satisfy the 2000psi requirement in the driver section but still produce undesirable shock strength with a low Mach number. Conversely, there are combinations, some not shown, that generate stronger shocks but require driver pressures that far exceed project parameters and requirements. These figures provide a direct comparison and way to balance structural feasibility against testing performance and project requirements.

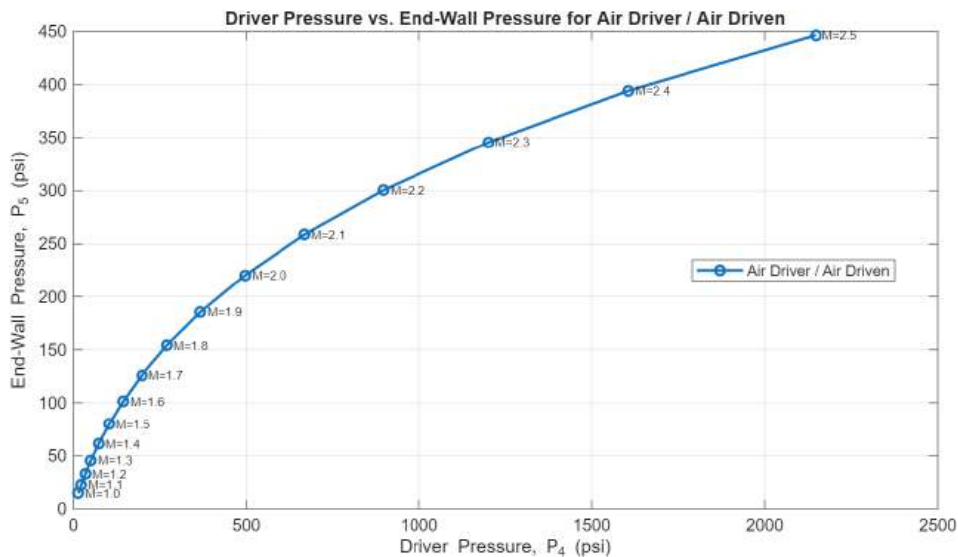


Figure 25 Driver Pressure Vs End Wall Pressure

Assumption and Note For Subsequent Calculations

Assumption: during a test the highest pressure within the system is the pressure within the driver section, all future calculations are made with this assumption. ^[1]

[1] Note: The team is performing more extensive research to validate this claim as we have encountered multiple viewpoints on this assumption with some researchers following our assumption where a few others like in source [26] are citing a massive pressure spike at the end wall.

Wall Thickness Calculation

A major factor in determining wall thickness is the requirement for the shock tube to safely contain high-pressure gas up to 2000 psi. Due to the material fatigue behavior under cyclic loading, a simple force analysis to get thickness was not sufficient in ensuring the safe holding of the high-pressure gas. Instead, the wall thickness was determined through a combination of fatigue analysis and pressure vessel theory, using Lamé's equation for thick-walled cylinders and plate bending theory for the end wall. The final radial wall thickness of the shock tube was evaluated to be **1.9702 in**. The end wall thickness of the face of the shock tube with intake and exhaust holes of 1/4 in was determined to be **1.6323 in**. An end wall thickness of **1.7183 in** will be used for the other wall to accommodate for the higher yield stress caused by test stage screw holes. The following calculations are for 2000 psi with a recommended max pressure of 500 psi and a factor of safety of 9.

The governing equation for determining the radial thickness of the tube, accounting for high-pressure gradients in thick-walled cylinders, is Lamé's Equation (Eq. 8).

$$t_{\text{shell}} = r_i \left(\sqrt{\frac{\sigma_{\text{allow}} + P_i}{\sigma_{\text{allow}} - P_i}} - 1 \right)$$

[8]

This expression defines the required wall thickness as a function of internal pressure P_i , inner radius r_i , and allowable stress σ_{allow} . The allowable stress is not taken directly from material properties but is instead derived through fatigue analysis to account for cyclic loading and the requirement for infinite life. The Marin Fatigue Factors used for analysis can be seen in eq. 9-14.

Commented [RR1]: Adjust calculations and numbers, include tables of design radii

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$$\begin{aligned}
K_a &= a (S_{ut} \cdot 10^{-3})^b \\
K_b &= \left(\frac{D_i}{0.3} \right)^{-0.107} \\
K_c &= 1 \\
K_d &= 0.9 \\
K_e &= 0.702 \\
K_{total} &= K_a K_b K_c K_d K_e
\end{aligned}
\tag{9-14}$$

K_a denotes the surface factor that reduces allowable stress through the presence of surface imperfections in the tube introduced by the surface finish. Assuming $a = 2.70$ for machined and polished surfaces and $b = -0.265$ for the shell being in bending, the value for K_a is 0.8599. K_b denotes the size factor and accounts for the reduction in fatigue strength with increasing component size. K_c , the loading factor, is assumed to be 1 in the case of bending due to the behavior of a pressure vessel. K_d , the temperature factor, is evaluated to be 0.9 for the worst-case scenario because there are instances in operation in which the temperature of the gases instantaneously reaches a high value due to the pressure. For a desired reliability of 99.99%, K_e , the reliability factor is 0.702. The total Marin Factor is then applied to the endurance limit of the selected material to simulate the actual limit under operating conditions as shown in eq. 15.

$$S'_e = S_e \cdot K_{total} \tag{15}$$

The ultimate tensile strength of the material alongside the modified endurance limit is substituted into eq. 16 to determine the maximum radial stress using the Goodman relation for fully reversed loading (infinite life).

$$\sigma_{\max, \text{radial}} = \frac{2}{\frac{1}{S'_e} + \frac{1}{S_{ut}}} \tag{16}$$

The tube end wall assumes a smooth constant geometry; however, that is untrue for the end wall with the presence of exhaust and intake holes. This introduces a large stress concentration at the location with varying geometry requiring additional correction. The stress concentration factor to account for the holes is displayed in eq. 17 where R_i is inner tube radius and d is hole diameter.

$$K_t = 1 + 2 \left(\frac{d}{R_i} \right) + 3 \left(\frac{d}{R_i} \right)^2 \quad [17]$$

Kt was then converted to fatigue stress concentration through eq. 18 to perform fatigue analysis on the end wall to determine its thickness.

$$K_f = 1 + q(K_t - 1) \quad [18]$$

The same modified endurance limit is used for the end wall since all desired Marin factors are unchanged between the end wall and tube wall. Considering the modified endurance limit, fatigue stress concentration, and ultimate tensile stress of the material, a modified version of the Goodman equation was used to determine allowable stress in the axial direction as displayed in eq. 19.

$$\sigma_{\max} = \frac{2}{\frac{1}{S'_e/K_f} + \frac{1}{S_{ut}}} \quad [19]$$

With the allowable stress, pressure rated for holding, inner radius, and safety factor of 9, eq. 8 was evaluated to determine the radial wall thickness. End wall thickness calculation uses the equation associated with plate bending theory as shown in eq. 20. The equation takes into account tube diameter, max allowable stress, working pressure (2000 psi), and a coefficient (C = 0.33) assuming a fully welded boundary.

$$t_{\text{end}} = d \cdot \sqrt{\frac{C \cdot P}{\sigma_{\max}}} \quad [20]$$

Figure 26 shows the mapped-out process for getting the radial thickness while Figure 27 presents the process for end wall thickness.

$$S'_e \rightarrow \text{Goodman} \rightarrow \sigma_{\max} \rightarrow t_{\text{tube}}$$

Figure 26 Radial Wall Thickness Roadmap

$$K_t \rightarrow K_f \rightarrow \text{Goodman} \rightarrow \sigma_{\max} \rightarrow t_{\text{end}}$$

Figure 27 End Wall Thickness Roadmap

Table 2 displays varying wall thickness dimensions for different operation pressures. This is provided in case there is a desire to produce specialized shock tubes at lower cost values specifically for pressures other than 2000 psi.

Table 2 Wall thickness dimensions for 500, 1000, and 2000 PSI

Driver Pressure (PSI)	Driver End Wall Thickness (in)	Driven End Wall Thickness (in)	Radial Thickness (in)
500	0.816	0.859	0.283
1000	1.154	1.215	0.644
2000	1.632	1.718	1.970

Diaphragm Thickness Calculation

The diaphragm thickness is the key component of making the shock tube work. The thickness for the shock tube is what determines the maximum pressure within the driver section. A thinner membrane means that the max pressure in the driver section will be less than that of a thicker membrane. The following equation is used to estimate the membrane thickness.

$$t = 0.275 \frac{pR\sqrt{E}}{\sigma_{fail}^{\frac{2}{3}}} \quad [21]$$

This equation contains the pressure difference, p , across the membrane, the unsupported radius of the membrane, R , Youngs modulus, E , and the effective burst stress σ_{fail} of the diaphragm material. In practice the burst stress of a material is not given as a material property. This value must be determined experimentally through multiple trials. The equation above can be rewritten into the following:

$$\sigma_{fail} = 0.423 \left(\frac{Ep^2R^2}{t^2} \right) \quad [22]$$

This equation should be used in combination with a simple experiment to determine the burst stress of the steel material. To start the experiment a set of steel diaphragms must be manufactured with known thicknesses, the team recommends using the ultimate tensile strength, UTS, of the material as an initial guess in the thickness equation. Then shock tube tests should be performed for each of those diaphragm thicknesses that were designed for certain pressures. The burst pressure results should be documented and used in equation 21 to back calculate the burst stress of the material and then the thickness can be more

accurately modeled. It should be noted that the pressure at which a diaphragm bursts can differ from diaphragm to diaphragm because factors like rate of pressure rise, manufacturing differences, and even material properties. [27][28]

Tube Length Calculation

There are many components used to determine the length of the driver and the driven sections of the tube. First is to make sure that the shock wave is fully planar when it impinges on the test specimen. Second is to make sure that the expansion waves don't catch up. Third is to make sure that the system fits inside the size constraints and leave room for wires and tubes and other required supports. The process used to determine the lengths is iterative since the length of the driven section is dependent on the length of the driver section.

For this design, the driver section length is chosen as a starting point. The length of the driven section is then based on the driver section length, the desired pressures, gas constants, and tube properties. Since this process is iterative to get a length that fits within the size constraints, a MATLAB script was made to automate the calculations. The first step is to solve for the shock Mach number.[29] This step is a redundant measure to confirm the previous pressure calculations and to attain a more accurate value. To find the Mach number we equate the speed of U2 and U3 which are variables representing flow velocities in the driven and the driver sections and solve for the Mach number. The equation is given below. (eq. 22) [30] [31]

$$\frac{2a_1}{\gamma_1 + 1} \left(\frac{M_s^2 - 1}{M_s} \right) = \frac{2a_4}{\gamma_4 - 1} \left[1 - \left(\left(1 + \frac{2\gamma_1}{\gamma_1 + 1} (M_s^2 - 1) \right) \frac{p_1}{p_4} \right)^{\frac{\gamma_4 - 1}{2\gamma_4}} \right] \quad [23]$$

Next the shock speed is determined by multiplying the area of the tube by the Mach number and the driven gas density is also determined. To make sure that the majority of the test specimen area is planar, the max distance before the boundary layer expands beyond a set percentage which is 1.5% of the radius for this system is found. The equation used is a viscous shock planarization length equation given below (eq 23).[32] Now to find the minimum distance needed for the shockwave to be fully planar, the speed of the shockwave times the length of the driver over the speed of sound in the driver is calculated.

$$x_{\text{planar,max}} = \frac{\epsilon^2 D^2 \rho_1 U_s}{25 \mu_1 C_r} \quad [24]$$

The driver length is recalculated using this driven length to make sure that the driver length provides adequate time for the expansion waves to propagate without interfering with the shockwave. Based on an initial length 0.914 m, the minimum driven length is found to be 1.831 m which is approximately 3 ft and 6ft respectively. This calculation assumes that the pressure is kept at a max value of 500 psi. To account for 2000 psi in the driver section, we need to repeat this process using the higher pressure which yields a needed length in the driven section of 7.5ft. For the internal length, to account for cases from 500 psi to 2000 psi, the driven section will be 8 ft which leaves one foot for end plates and wiring.

It should be noted that the team does recommend increasing the amount of space available for a shock tube. Most tubes are 20-30ft rather than 12ft. The increased length will allow for more consistent and repeatable results and it will allow for more design freedom for the tube.

Bolt Calculation

The total applied load on the flange joint was calculated as 251.54 kN, based on the internal pressure of 500 psi acting across the full flange face area. A safety factor of 2.5 was applied to account for dynamic loading and joint separation requirements, giving a design load of 628.86 kN. With 8 bolts equally spaced on a 177.8 mm bolt circle, the load per bolt is:

$$F_{bolt} = \frac{628.86}{8} = 78.61 \text{ kN per bolt}$$
 [25]

Based on this per-bolt load, M14 Grade 10.9 bolts were selected. Grade 10.9 bolts have a proof strength of 830 MPa per ISO 898-1 [33], giving a proof load of 95.45 kN for an M14 bolt. The bolt utilization is therefore:

$$\text{Utilization} = \frac{78.61}{95.45} = 82.3\%$$
 [26]

This falls within the accepted range for structural bolted joints [34]. The required tightening torque per ISO 898-1 is 179 Nm. The complete bolt selection parameters are summarized in Table 3.

Table 3 Bolt Load Parameters

Parameter	Value
Applied Load	251.54 kN
Safety Factor	2.5
Design Load	628.86 kN
Bolt Circle (PCD)	177.8 mm (7")
No. of Bolts	8

Load per Bolt	78.61 kN
Selected Fastener	8× M14 Grade 10.9
Proof Load	95.45 kN
Utilization	82.3%
Tightening Torque	179 Nm
Bolt Standard	ISO 898-1

SOLIDWORKS FEA Analysis

A static structural finite element analysis was performed in SolidWorks Simulation on the driver section, as it represents the highest-pressure component in the assembly, subject to the full 500 psi driver pressure p4. The driven section experiences lower incident pressure and was therefore not the governing case for structural analysis.

Setup and Boundary Conditions

The driver section was modeled as a hollow cylinder with a 4" outer diameter, 3" inner diameter, and 1" wall thickness. The material was assigned as AISI 304 stainless steel with a yield strength of $2.068 \times 10^8 \text{ N/m}^2$ (30,000 psi) and elastic modulus of $1.93 \times 10^{11} \text{ N/m}^2$. A uniform internal pressure of 500 psi ($3.447 \times 10^6 \text{ N/m}^2$) was applied to the inner wall surface. A fixed boundary condition was applied at one end of the tube, constraining all translational and rotational degrees of freedom. A static nodal stress analysis was performed with a normal mesh deformation scale of to visually amplify displacement results.

Results

The von Mises stress distribution across the driver section is shown in Figure 28. The results are summarized in Table 4 below.

Table 4 SOLIDWORKS FEA Analysis Results

Parameter	Value
Maximum von Mises Stress	$3.690 \times 10^6 \text{ N/m}^2$ (535 psi)
Minimum von Mises Stress	$1.562 \times 10^6 \text{ N/m}^2$ (227 psi)
Yield Strength (AISI 304)	$2.068 \times 10^8 \text{ N/m}^2$ (30,000 psi)

The tube body exhibits uniformly low stress throughout, shown by the green contour across the full length of the cylinder. Stress concentrations are observed at the end faces, shown in orange and red. These concentrations are a direct artifact of the fixed boundary condition applied at the tube end and are not representative of stress in the tube wall under operational loading. In a fully assembled configuration with flange collars and bolted connections at both ends, these end effects would be distributed across the joint.

The maximum stress of 535 psi is 56 times below the yield strength of AISI 304 stainless steel, confirming that the 1" wall thickness provides substantial structural margin at the 500psi operating pressure. These results are consistent with the hand calculations performed using the Lamé thick-wall pressure vessel equation, which yielded a similarly large safety margin for this tube geometry and pressure condition.

The FEA analysis was performed under static, isothermal conditions and does not account for transient temperature effects during operation. When the diaphragm ruptures and the shock wave propagates, the gas temperature rises significantly behind both the incident and reflected shocks. These elevated temperatures would locally reduce material yield strength and introduce thermal stresses across the tube wall due to the temperature gradient between the inner and outer surfaces. Repeated pressurization cycles would also introduce thermal fatigue not captured in a static analysis. Given the large factor of safety observed in the FEA results, these effects are unlikely to be critical, but a coupled thermal-structural analysis would be required for full characterization [35][36].

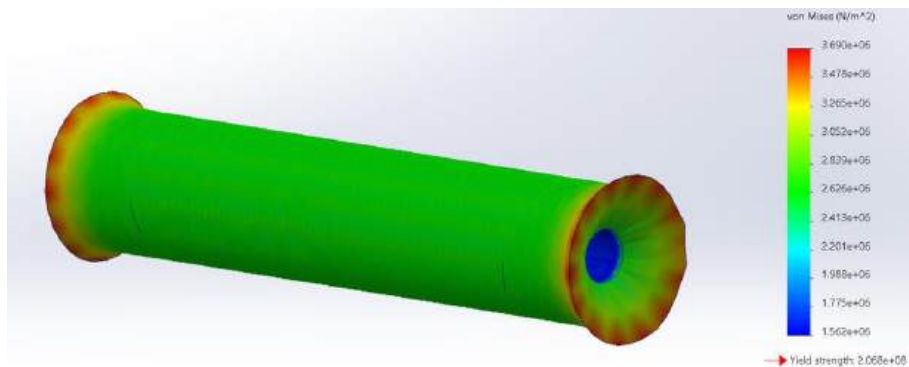


Figure 28 SOLIDWORKS FEA Analysis of Driver Section

Mockup and Prototyping

The goal of the project is to design and validate a modular shock tube test platform without the expectation of a physically manufactured prototype. Thus, there will be no prototypes of the shock tube by the end of the project; however, in place of a prototype is a well-designed model with experimentally validated dimensions to be used for future development. The design will follow codes and standards laid out by already established organizations for safety reasons. The design will adhere to the design constraints imposed by ASME standards for pressure vessel design. To build a final physical prototype, it is required that guidelines for machinery operation from OSHA and maintenance from API be

upheld during the lifespan of the device. Given many more safety constraints in terms of the handling of high-pressure vessels, a physical prototype is not feasible for a one semester project; thus, an experimentally validated and digitally designed final version that adhere to the safety standards will be refined.

Risk Assessment, Safety and Liability

The proposed design introduces several safety concerns due to the use of compressed gas, rapid diaphragm rupture, high system pressures, and potential release of stored energy. To evaluate these concerns in a formalized way, a risk assessment approach based on severity, likelihood, and detectability was used. The highest priority risks were associated with pressure vessel failure, uncontrolled diaphragm rupture, gas leakage, projectile or fragment hazards, and operator exposure during pressurization and firing. The following Table 4 organizes the teams findings.

Table 5 Simplified Risk Assessment for Shock Tube Operation

Failure Mode	Consequence	Risk Level	Mitigation
Over pressurization of driver section	Structural rupture or sudden release of stored energy	High	Use conservative operating pressure limits, pressure-rated components, regulator control, pressure gauges, and pressure relief protection.
Unexpected diaphragm rupture	Premature shock event, equipment damage, or operator exposure	High	Select diaphragm thickness and material carefully, use repeatable clamping, pressurize gradually, and maintain a clear exclusion zone.
Diaphragm or fitting fragmentation	Debris hazard or internal tube damage	Medium	Use ductile diaphragm materials, contain rupture within the membrane section, inspect parts after each test, and use operator shielding.
Gas leakage at seals or fittings	Loss of pressure, inconsistent results, or hazardous release of gas	Medium	Perform leak checks, inspect O-rings/gaskets, use compatible fittings, and verify proper tightening before operation.
Threaded or bolted joint failure	Joint separation, projectile hazard, or rapid depressurization	High	Size fasteners conservatively, verify thread engagement, apply proper torque, and

			inspect joints before each test.
Operator error during setup or firing	Unsafe pressurization, misfire, or exposure to hazardous areas	High	Use a written operating procedure, pre test checklist, and training

The most significant risk is the release of gas from the driver section. Because the shock tube relies on a pressure differential across a membrane, the driver section must safely contain the pressure before the diaphragm ruptures. This risk directly influenced the use of conservative pressure limits, and a high factor of safety throughout the design process.

The diaphragm was another major safety concern because it is intentionally designed to fail. This creates a unique design challenge in which the diaphragm must rupture reliably during testing at a specified pressure, but the surrounding tube structure must not fail. To reduce this risk, the diaphragm was treated as a controlled failure element. Its thickness, material properties, and knife edge contact zone were selected to promote rupture at the desired pressure. This approach reduces the likelihood of uncontrolled rupture elsewhere in the system and improves repeatability between tests.

Societal, Environmental and Sustainability Considerations

Shock tube design and operation must account for specific societal impacts, primarily concerning the safety of the personnel responsible for operating this device. Shock tubes, like this one, are dangerous pressure vessels that can store upwards of thousands of psi. This introduces a significant risk of structural failure, projectile hazards, compressed gas, toxic gas, and loud noise. As a result, the most important objective is safety for those operators. This design incorporates conservative pressure containment analysis, controlled firing procedures, intuitive use, and various control safety measures to ensure operator safety is preserved. These measures are paramount in protecting operators but also ensuring shock tube test repeatability within a shared laboratory environment.

Environmental considerations within shock tube design are centered around gas use, noise, and waste generation. Shock tubes can consume a significant amount of pressurized gases during testing; depending on gas choice outlined in the pressure region calculation section some of these gases can create hazards for workers and for the environment if handled improperly. For example, argon exposure can create an oxygen deficient environment and cause symptoms such as dizziness, headache, or asphyxiation in confined spaces [37]. These shock tube tests can also produce significant bursts of noise, potentially impacting on workers or the surrounding environment. Lastly, shock tubes produce waste as the metal diaphragms are not reusable and need to be exchanged

after every test, proper disposal of these testing materials is important for environment preservation.

Summary of Future Work

The team has continued to refine and finalize the shock tube design throughout the final phase of the project by further developing the pressure vessel design, validating critical components, and ensuring that the final recommendations satisfy customer requirements and safety considerations. The team has also performed research on pressure region calculations within the shock tube, FEA analysis standards for wall thicknesses in the tube and at the end walls, the nuances of the tube length, bolt calculations, material selection, diaphragm design, and end wall design. The team finalized the tube length, including an 8' driven section and 3' driver section, leaving 1' of space for any additional piping or tubing that needs to attach to the shock tube. The team also updated the design basis from an initial recommended maximum pressure test of 500 psi to a 2,000 psi design pressure in order to provide a greater safety margin and better align the design with potential operating conditions. Based on this updated design requirement, the team finalized the initial end wall thickness and pipe thickness, performed supporting calculations, and validated the design using SOLIDWORKS stress analysis modeling. These end wall and pipe thicknesses came out to 1" thick all the way around. The team also performed region pressure analysis to better understand the maximum pressures within the tube for the driver, driven, and testing sections, as well as to evaluate what gas combinations could be used to satisfy customer requirements and analyze how the shock front varies with Mach number. In addition, the team developed a step-by-step process for determining the required diaphragm thickness, which supports future testing and diaphragm selection. The team has also finalized the design of the intake and exhaust system and its components, ensuring that the hardware used for pressurization is integrated with the overall shock tube design and testing procedure.

As the project concludes, the team recommends that future work focus on experimental validation, final manufacturing review, and continued refinement of the diaphragm and end wall design. This includes physically testing diaphragm burst behavior, confirming manufacturing tolerances, reviewing sealing methods, and further evaluating the ease of use and safety of the end wall access procedure. This is fundamentally a safety-driven project, and the team is committed to ensuring that all final designs, specifications, and recommendations are accurate, practical, and safe for future use. The team has also prepared final presentation materials to share the project's design process, findings, and excitement toward the development of the shock tube. All final deliverables and remaining recommended next steps are depicted in the Gantt chart in Figure 31 and Table 6.

Team Contributions

Ryland: Edited Report, wrote intro, Existing products, Customer Requirements, Pressure Region Calculations, Societal and Environmental impact, and summary of future work sections.

Eric: Edited Report, wrote Design concept ideation, concept selection and justification, industrial design, and tube length calculation sections

Leo: Edited Report, wrote codes and standards, wall thickness calculation, air intake and exhaust system design, and mockup and prototyping section, also created Gantt chart figures

Daanish: Edited Report, wrote market research, design concept ideation, bolt calculation, FEA analysis, sections and did all SolidWorks modeling

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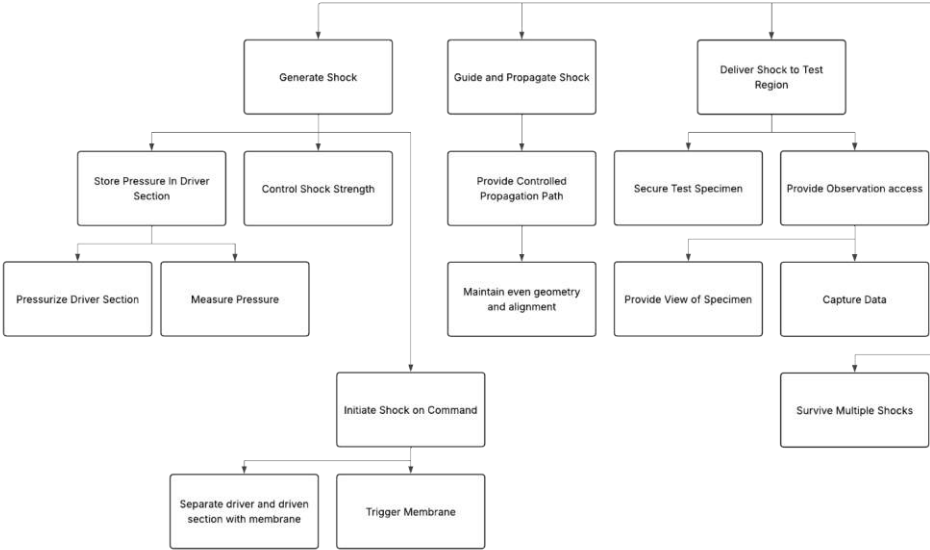
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Appendix

Impl



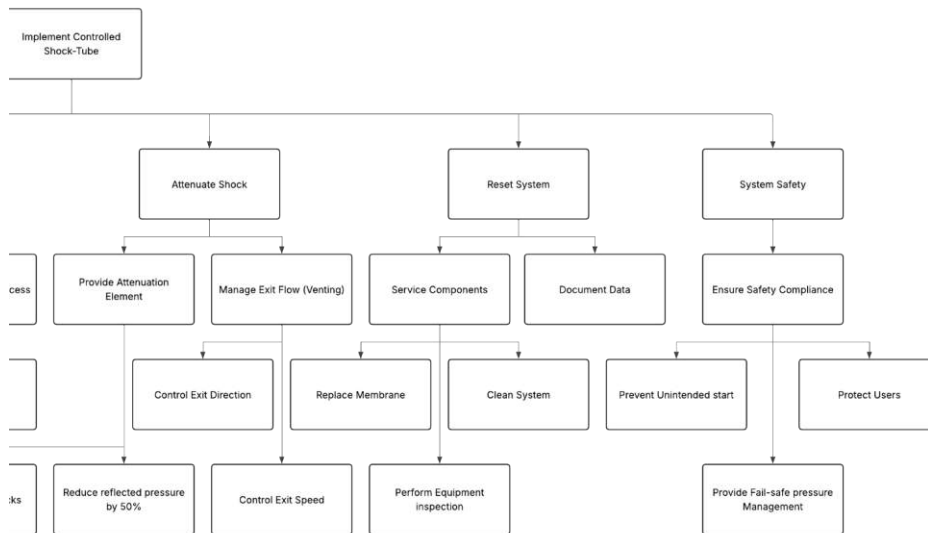
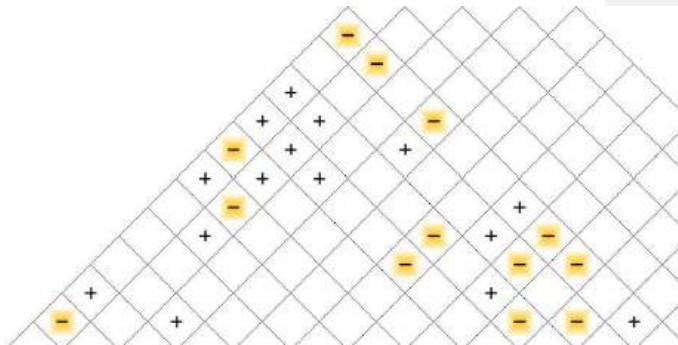


Figure 29 Function Tree



		Column.#											
		1	2	3	4	5	6	7	8	9	10	11	12
		Direction of Improvement											
Weight	Customer Requirements (Explicit and Implicit)												
		Range Of Pressure	Max Size	Min Cycle Time	Quick Access door	Set pressure value	Absorption section	Industrial available parts	Size of window	Time to vent	Max sound produced	Follows ASME	Lifecycle
10	Range of Pressure	●	○			○				▽	▽	▽	▽
10	Size Constraint	○	●			○	▽		▽	▽		▽	
8	Fast Cycle Time			●	○	○				▽		▽	
7	Ease of use		▽	○	●	●		▽	○	▽	▽	▽	
6	Reduced Reflection Pressure	○					●				▽	▽	
5	Standard Parts							●				●	●
3	Viewing window	▽	○						●			▽	
10	Venting System	▽	▽	▽						●		▽	
2	Sound Proof	▽					○			●	●	▽	
7	Safety Features	○			○	▽	▽	○		○	●	●	●
9	Doesn't break	○				▽	▽	○	●			▽	●

Target	5-2000psi	12' x 8ft'	<20min	Opens <1min	Control pressure <30s	Reduced by 50%	min 50% of system	min radius of 2"	Vents less than 10s	<85db	Yes	10000000 cycles
Max Relationship	9	9	9	9	9	9	9	9	9	9	9	9
Technical Importance Rating	201	146	103	108	163	86	100	139	164	104	173	199
Relative Weight	12%	9%	6%	6%	10%	5%	6%	8%	10%	6%	10%	12%
Weight Chart												

Figure 30 House of Quality

Table 6 List of tasks needed to be completed for the semester that is used for the Gantt chart.

WBS NO.	Task Name	Status	Assigned To	Start Date	End Date	Duration in days	Milestone
1	Major Deadlines and Assignments	In Progress		02/07	05/01	85	Yes
1.1	Report 1	Complete	Team	02/07	02/09	2	Yes
1.2	Report 2	Complete	Team	03/12	03/20	8	Yes
1.3	Report 3	Not Started	Team	04/25	05/01	7	Yes
1.4	EXPOSITION	Not Started	Team	04/15	05/01	17	Yes
2	Project Conception and Initiation	Complete		01/20	01/28	9	No
2.1	Project Assignment	Complete	Galech	01/20	01/20	1	No
2.2	Sponsor Consultation	Complete	Team	01/22	01/22	1	No
2.3	Problem Understanding	Complete	Eric	01/23	01/28	6	Yes
3	Preliminary Design	In Progress		01/25	04/13	80	No
3.1	Research	Complete	Team	01/25	02/28	35	No
3.2	Preliminary Drawings	Complete	Eric	01/25	03/06	25	Yes
3.3	Calculation on Preliminary Drawings	Complete	Team	02/18	03/11	23	No
3.4	Design Validation	In Progress	Leo	03/15	04/03	29	No
4	Project Finalization	In Progress		02/16	04/15	60	No
4.1	Choose Finalized Design	Complete	Team	03/02	03/11	10	Yes
4.2	Testing	In Progress	Team	03/15	04/05	31	No
4.3	CAD Modeling	In Progress	Team	03/23	04/05	53	No
4.4	CFD Analysis	In Progress	Deanish	03/15	03/25	39	No
4.5	FEA Analysis	In Progress	Plyland	03/15	03/25	39	No

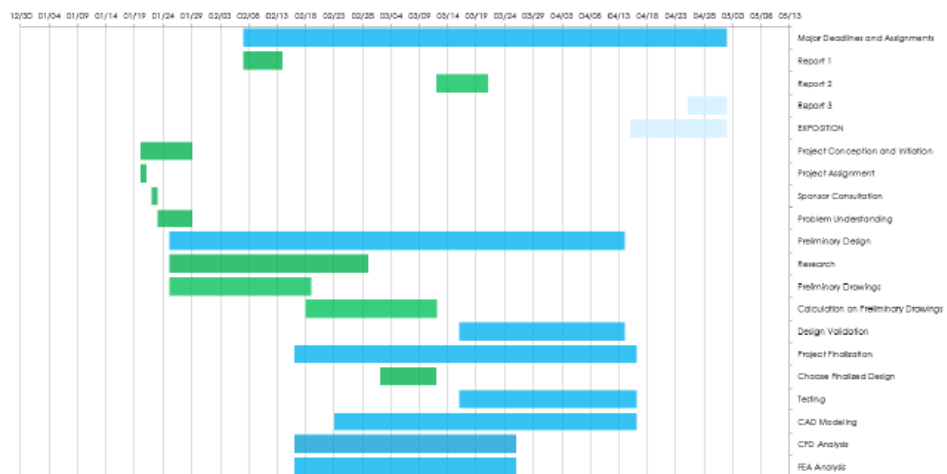


Figure 31 Gantt chart depicting the teams predicted progress for the semester.